

Google Cloud Functions

Event-driven, serverless compute - from trigger to code in milliseconds

Matthew • CS Presentation

Agenda

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Intro

What Cloud Functions are, and why they exist

02

History

From 2017 beta to today’s 2nd-gen platform

03

Demo

Deploying and triggering a live function

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Capstone Usage

Where this fits into your own capstone project

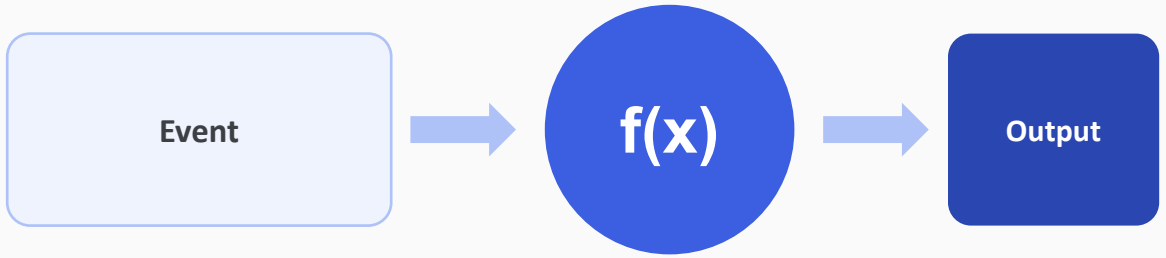
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Summary

Key takeaways to remember

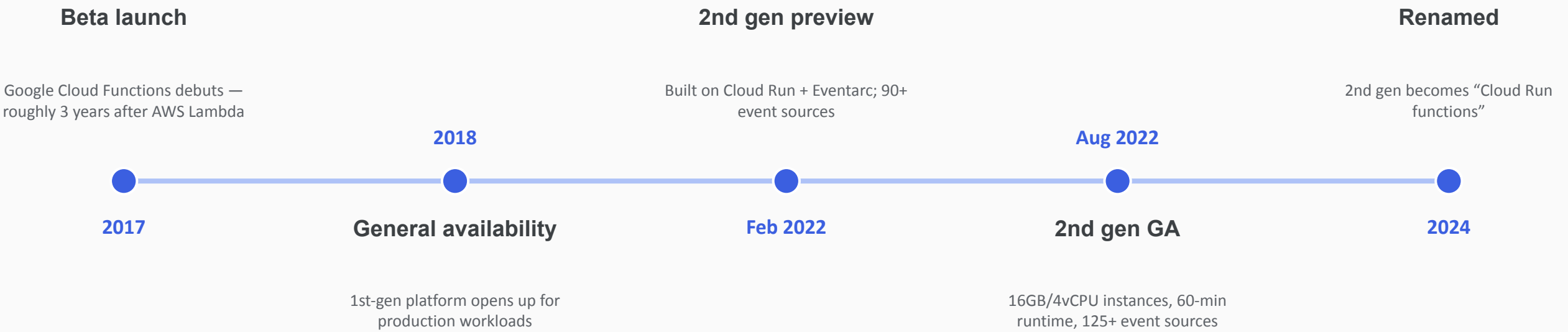
Intro: What & Why

- Serverless: no servers to provision, patch, or scale by hand
- Event-driven: code runs in response to triggers: HTTP calls, file uploads
- Pay-per-use: billed only for actual execution time, down to the millisecond
- Auto-scaling: from zero instances to thousands, automatically



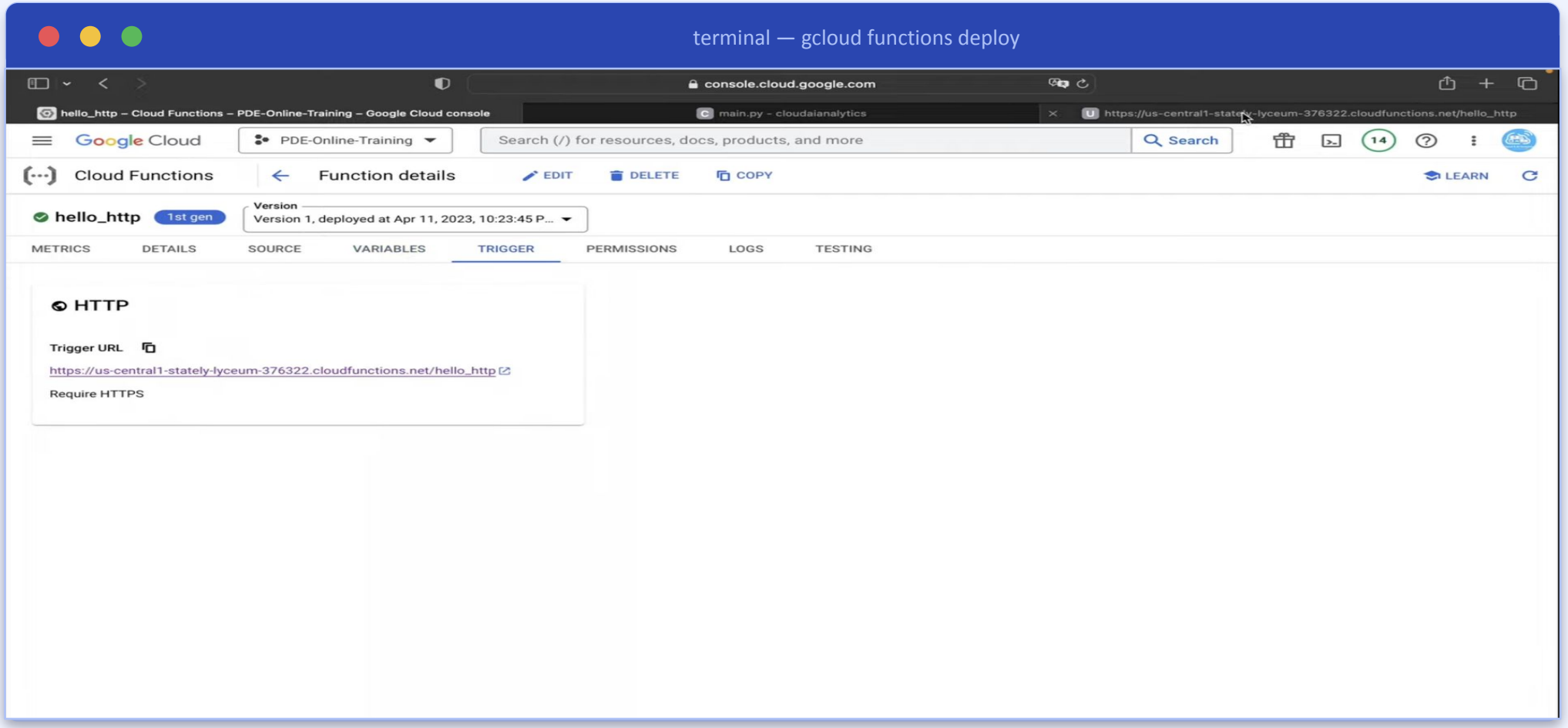
You write the function. Google runs it, scales it, and tears it down.

History


Beta launch
Google Cloud Functions debuts — roughly 3 years after AWS Lambda
2017
2018
General availability
1st-gen platform opens up for production workloads
Feb 2022
2nd gen preview
Built on Cloud Run + Eventarc; 90+ event sources
Aug 2022
2nd gen GA
16GB/4vCPU instances, 60-min runtime, 125+ event sources
2024
Renamed
2nd gen becomes “Cloud Run functions”

Demo

Deploying and triggering a live HTTP function



Potential Usage in Capstone Projects

1 API & webhook backends

Handle form submissions or third-party webhooks without standing up a server

2 Event-driven data pipelines

Trigger on a Cloud Storage upload or Pub/Sub message to process data as it arrives

3 Scheduled automation

Run nightly jobs — cleanup, reports, syncs — on a Cloud Scheduler cron

4 Lightweight ML inference

Wrap a trained model behind an HTTP endpoint for on-demand predictions

Summary

- Cloud Functions let you ship code without managing infrastructure
- You pay only for what actually runs — down to milliseconds
- 2nd gen adds bigger instances, longer runtimes, and 125+ triggers
- A strong fit anywhere your capstone needs a small, event-driven piece of logic

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servers to manage

Scale to zero when idle, scale to thousands on demand

Thank You